Program: B.Sc. – Information Technology (2026 - 27)		Semester: VI	
Course: Advanced Artificial Intelligence LAB		**Course Code: UIAAI354P**	
Teaching Scheme		**Evaluation Scheme**	
Lecture (Hours per week)	**Credit**	**Continuous Assessment (CA) (Marks -20)**	**Semester End Examinations (SEE) (Marks- 30 in Question Paper)**
2	1	20	30

Learning Objectives:

- To understand the concepts of soft computing techniques including neural networks and fuzzy logic.
- To learn the design and implementation of basic and advanced neural network models for pattern recognition and classification.
- To gain practical experience in training neural networks using learning algorithms such as perceptron learning and backpropagation.
- To explore advanced AI models such as Hopfield Networks and Gaussian Mixture Models for clustering and associative memory.
- To develop practical skills in Natural Language Processing (NLP) including text preprocessing, N-gram models, and parsing techniques.

Course Outcomes:

- Students will be able to implement basic neural network models such as Perceptron, ADALINE, and Multi-Layer Perceptron.
- Students will be able to design and apply fuzzy logic systems for solving real-world control problems.
- Students will be able to implement advanced neural network techniques such as Hopfield Networks and Gaussian Mixture Models.
- Students will be able to apply NLP techniques including text preprocessing, N-gram modeling, and syntactic parsing.
- Students will be able to analyze and evaluate the performance of different AI models for classification, clustering, and language processing tasks.

Outline of Syllabus:		
Topics		**Duration (Lecture)**
Module I	**Soft Computing & Basic Neural Networks** **Practical 1: Implementation of Simple Perceptron** • Implement binary classification using Perceptron. • Train on AND/OR logic gates dataset. • Plot decision boundary. **Practical 2: ADALINE and MADALINE Network** • Implement Adaptive Linear Neuron (ADALINE). • Train using gradient descent. • Compare Perceptron vs ADALINE performance. **Practical 3: Backpropagation Network (Feedforward Neural Network)** • Implement Multi-Layer Perceptron (MLP). • Train using backpropagation algorithm. • Experiment with different activation functions (Sigmoid, ReLU, Tanh). • Analyze learning curves. **Practical 4: Fuzzy Logic System Design** • Design a simple fuzzy inference system. • Define membership functions. • Implement fuzzification, rule evaluation, defuzzification. • Example: Temperature control system.	
Module II	**Advanced Neural Network Models** **Practical 5: Hopfield Network Implementation** • Implement Hopfield Network. • Store and retrieve binary patterns. • Demonstrate associative memory behavior. • Analyze energy convergence. **Practical 6: Gaussian Mixture Model (GMM)** • Implement GMM using Expectation-Maximization (EM) algorithm. • Apply to clustering dataset. • Compare with K-means clustering.	

Module III:	**Natural Language Processing** **Practical 7: Text Preprocessing and N-gram Model** • Perform tokenization, stop-word removal, stemming/lemmatization. • Generate Unigram, Bigram, Trigram models. • Calculate probabilities of word sequences. **Practical 8: NLP Parsing** • Perform Constituency Parsing using CFG. • Perform Dependency Parsing using spaCy. • Implement simple Probabilistic Context-Free Grammar (PCFG). • Compare parse trees for ambiguous sentences.	
Total Lectures		**30**

Books and References:

Sr. No.	Title	Author/s	Publisher	Edition	Year
1.	Neural Networks,A Fuzzy Logic and Genetic Algorithms: Synthesis and Applications	S. Rajasekaran & G. A. Vijayalakshmi Pai	PHI Learning	1st	2003
2.	Speech and Language Processing	Daniel Jurafsky & James H. Martin	Pearson / Prentice Hall	2nd Edition	2014
3.	Principles of Soft Computing	S. N. Sivanandam & S. N. Deepa	Wiley-India	2nd Edition	2007
4.	Artificial Neural Networks: A Practical Course	Ivan Nunes da Silva, Danilo Hernane Spatti, Rogerio A. Flauzino, Luisa H. B. Liboni, Silas F. dos Reis Alves	Springer	1st Edition	2016
5.	Neural Networks and Deep Learning: A Textbook	Charu C. Aggarwal	Springer	1st Edition	2018